Ahan Bhatt

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Summary

Recent M.S. graduate in Computer Science and Engineering with experience building transformer compilers, GPU-accelerated LLM inference systems, adversarial evaluation infrastructure, and trustworthy AI platforms. Strong background in Python, C++, CUDA, Triton, PyTorch, FastAPI, and full-stack ML systems, supported by research experience and publications across LLM evaluation, explainable AI, time-series modeling, and applied machine learning.

Education

University of California, Santa Cruz [3.82 GPA] — Santa Cruz, California
Master of Science in Computer Science and Engineering — *Aug. 2025 – Jun 2026*

Indus Institute of Technology and Engineering [3.94 GPA] — Ahmedabad, India
Bachelor of Technology in Computer Engineering — *Aug. 2020 – July 2024*

Technical Skills

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS, R
Machine Learning: PyTorch, TensorFlow, Keras, Transformers, scikit-learn, NumPy, Pandas, Matplotlib
ML Systems and Compilers: CUDA, Triton, C++20, pybind11, custom IRs, operator fusion, kernel selection, memory/execution planning, paged KV caching, quantization
LLM and Data Tools: LangChain, ChromaDB, RAG, vector databases, PyPDF
Backend and Deployment: Node.js, FastAPI, Flask, REST APIs, Docker, Kubernetes, Linux, Git, CI/CD, Nginx
Cloud: AWS EC2, S3, IAM, Lambda, SageMaker, CloudWatch
Data Engineering: PostgreSQL, Redis, ETL, schema design
Software Engineering: TDD, API integration, DSA, system design

Experience

Researcher — September 2025 – June 2026
AI Explainability and Accountability (AIEA) Lab @ UCSC — *Santa Cruz, California*
- Built ReliaGuard Studio and BreakPoint with Next.js, TypeScript, FastAPI, PostgreSQL, Docker, and MLflow to detect over/underreliance in human-AI decisions and compile real LLM failure modes into adversarial eval suites using neuro-symbolic modeling, conformal gating, multi-judge validation, and explainable dashboards.

Data Scientist — June 2024 – May 2025
AI Institute of South Carolina — *Columbia, South Carolina (remote)*
- Conducted research on multimodal toxicity mitigation and anomaly detection, using PostgreSQL-backed experiment tracking to manage dataset metadata, model outputs, evaluation metrics, and benchmarking results for projects like DE-HATE and Time Series Foundation Model evaluation.

Data Scientist — Jan. 2024 – May. 2024
Space Applications Centre, ISRO — *Ahmedabad, Gujarat*
- Built an end-to-end LSTM satellite clock-bias prediction platform with Apache Kafka live streaming and a Next.js/React/TypeScript UI, automating preprocessing and anomaly filtering to improve navigation timing forecasts.

Software Engineer (AI) — May 2022 – Jan. 2024
Indus Institute of Technology — *Ahmedabad, Gujarat*
- Developed Node.js-backed LLM pipelines for a university website chatbot, automated knowledge graph generation, and retrieval-augmented medical QA, evaluating GPT-4, LLaMA-2, and BERT for structured semantic extraction and grounded reasoning.

Projects

CacheIR **[CUDA, Triton, C++20, custom compiler IRs, PyTorch, pybind11, FastAPI, NumPy]**
- A narrow, inspectable compiler and runtime for GPU-accelerated decoder-only transformer inference. It imports Llama/Mistral/Qwen-style models, lowers them into a custom IR, applies transformer-specific optimizations, plans memory and execution, selects CUDA kernels, and runs inference through a reference runtime with vLLM-compatible serving concepts.

BreakPoint **[LLM evaluation pipelines, multi-judge validation, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, MLflow, DuckDB]**
- A failure-to-eval compiler for LLM systems that mines real failure modes into versioned adversarial regression suites with DSL-defined traps/rubrics, multi-judge calibration, RAG/tool simulators, CI gates, FastAPI/Next.js dashboards, and 360 validated eval cases across 9 failure families.

ReliaGuard Studio **[Neuro-symbolic AI, conformal prediction, FastAPI, Next.js, TypeScript, PostgreSQL, MLflow, Docker, scikit-learn]**

Research Forge – A Self-Improving Research Agent **[LangGraph, autonomous agents, Neo4j, OpenAI API, Pydantic, Streamlit, Python]**

Meeting AI Assistant [**Real-time multimodal AI, Electron, React, OpenAI API, Node.js, Express, TypeScript**]

Time Series Foundation Model Evaluation for Anomaly Detection [**TimeGPT, Chronos, Time-MOE, PyTorch, TensorFlow, scikit-learn, Pandas, Python**]

Knowledge Graph Generation from Large Language Models [**GraphRAG, GPT-4, LLaMA-2, BERT, Transformers, Python**]

Med-Bot: Retrieval Augmented Medical Assistant [**Llama-2, AutoGPTQ, RAG, LangChain, ChromaDB, PyTorch, Flask, PyPDF, Python**]

Publications

Leveraging LSTM for Predictive Modeling of Satellite Clock Bias

- *Presented at CDMA 2025 conference and available on IEEE Xplore*

Generating Knowledge Graphs from Large Language Models

Time Series Foundational Models: Their Role in Anomaly Detection and Prediction

- *Presented at AAAI 2025 Anomaly Detection in Scientific Domains Workshop*

Med-Bot: An AI-Powered Assistant to Provide Accurate and Reliable Medical Information